

Mukesh Balaji R

Software Engineer | Frontend Engineer (Angular | React)

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PROFESSIONAL SUMMARY

Frontend Software Engineer specializing in Angular, TypeScript, and enterprise web applications, with experience building scalable frontend architectures, reusable UI systems, and high-performance visualization frameworks.

Architected and currently own an Angular UI Component Library with 80+ reusable components, developed a custom HTML Canvas Charting Library from scratch, and built AI-powered automation solutions using OpenAI APIs to streamline frontend development workflows.

Proven track record of delivering accessible, maintainable, and scalable solutions across multiple US Government and enterprise platforms while driving frontend modernization initiatives using Angular Signals, RxJS, and modern engineering best practices.

PROFESSIONAL EXPERIENCE

Software Engineer Frontend

Oct 2024 – Present

Protechsoft Technologies Pvt. Ltd

Chennai

- Developed and maintained enterprise-scale Angular applications for multiple US Government platforms, including workforce management, career development, and child support systems serving large-scale public sector operations.
- Architected, developed, and currently serve as the **primary owner** of a custom Angular UI Component Library comprising **80+ reusable and configurable components**, improving development consistency and accelerating frontend delivery across projects.
- Designed and developed a **custom HTML Canvas Charting Library** from scratch, delivering advanced and highly configurable visualizations beyond traditional charting solutions.
- Built an **LLM-powered automation solution** using OpenAI APIs that generates chart configurations from images, significantly reducing manual development effort and improving productivity.
- Led Angular modernization initiatives by migrating legacy state management patterns to Angular Signals, improving application reactivity, reducing complexity, and enhancing maintainability.
- Integrated and optimized complex REST API workflows using **RxJS** and reactive programming patterns for efficient state and asynchronous data management.
- Developed responsive and accessible interfaces aligned with enterprise and government accessibility standards including **WCAG 2.1 AA**.
- Recognized with **two merit-based salary increments within a year** for exceptional performance, technical ownership, and delivery of high-impact engineering initiatives.
- Collaborated with cross-functional teams in agile environments to deliver scalable, maintainable, and enterprise-grade frontend solutions.

PROJECTS

1. Enterprise UI Component Library | Angular, TypeScript, SCSS

- Designed and developed a centralized Angular component ecosystem with **80+ reusable components**.
- Implemented flexible theming using CSS Variables, extensive customization capabilities, and comprehensive developer documentation.

- Served as the primary owner responsible for architecture, maintenance, enhancements, and adoption.

2. Custom HTML Canvas Charting Library | Angular, TypeScript, HTML Canvas

- Built a high-performance chart rendering framework from scratch using HTML Canvas.
- Developed 15+ customizable chart types with advanced interactions and enterprise-grade flexibility.
- Served as the primary owner responsible for architecture, maintenance, enhancements, and adoption.

3. LLM-Powered Chart Configuration Generator | OpenAI API, Angular, TypeScript

- Developed an AI-powered solution that converts chart images into chart configuration objects.
- Leveraged OpenAI APIs and prompt engineering techniques to automate frontend workflows.

4. WIOA - US Government Workforce Management Platform | Angular Signals, TypeScript, RxJS

- Developed scalable Angular modules supporting workforce administration, compliance reporting, and operational workflows across a large-scale government platform.
- Implemented responsive and accessible interfaces following government standards.

5. CareerStart.ai - US Government Career Development Platform | Angular Signals, TypeScript, RxJS

- Developed career exploration and workforce readiness features for a US Government platform serving students and job seekers.
- Led Angular Signals migration initiatives, modernizing application architecture and improving maintainability and reactivity.

6. NJ Kids - US Government Child Support System | Angular Signals, TypeScript, RxJS

- Developed scalable Angular modules supporting child support case-management workflows and government business processes.
- Built reusable forms, validation systems, and data-driven interfaces to improve consistency and maintainability.

TECHNICAL SKILLS

Frontend Technologies: Angular, React, TypeScript, JavaScript (ES6+)

Reactive Programming & State Management: RxJS, Angular Signals, Redux

UI & Styling: HTML5, CSS3, SCSS, Tailwind CSS, PrimeFlex

Component & Design Systems: PrimeNG, Ant Design, Custom Component Library Development

Data Visualization: HTML Canvas, Chart.js, Custom Chart Framework Development

APIs & Integration: REST APIs, OpenAI API Integration

Tools: Git, GitHub, Postman

Other: Prompt Engineering, AI-assisted Development

EDUCATION

Bachelor of Technology in Chemical Engineering

Dr. M.G.R Educational and Research Institute

2021

Chennai

Web Development & React JS

Softlogic Academy

2023

Chennai